



LOGAN

HYPERSONIC VEHICLE MODEL

CHEAT SHEET

FROM CONCEPT
TO SIMULATION

FLIGHT DYNAMICS | SCRAMJET | GNC | MISSION SIMULATION | FLIGHTGEAR

1 PREREQUISITES

Software versions used for development and testing:

Software	Version
MATLAB	R2024b
Simulink	R2024b
Aerospace Toolbox	R2024b
FlightGear	2020.3.17

i The model is expected to work with newer MATLAB versions, provided that the Simulink model is converted to the corresponding version.

2 GET THE MODEL



Clone the repository

```
git clone https://github.com/ermersonmoura/  
LOGAN-Hypersonic-Vehicle-Model.git
```



Open in MATLAB

```
cd LOGAN-Hypersonic-Vehicle-Model
```



Open the model

```
open('logan.slx')
```

3 SIMULATION WORKFLOW

- 1 Configure simulation settings in `logan_init.m`
- 2 Select model options, autopilot modes and references
- 3 Run the simulation
- 4 Visualize and analyze results
- 5 (Optional) Enable FlightGear for real-time visualization

4 MODEL FLAGS AND OPTIONS

General configuration flags in `logan_init.m`.

Flag	Default	Description
Manual_Control	0	Enables manual control using a joystick. Set to 1 and uncomment the manual control block in Simulink.
Trimming	0	Enables the trimming procedure (set to 1).
Fuel_Freeze	0	Freezes fuel consumption (set to 1).
TECS_ON	0	Enables the Total Energy Control System (TECS) (set to 1).
PA_TECSEngineInput	0	Provides acceleration/deceleration reference to the TECS loop (set to 1).

i All flags are set to 0 by default. Enable only the features you need for your simulation.

5 ENGINE CONTROL

Engine control configuration.

Variable	Default	Description
PA_EngineMode	0	Mach-tracking control mode. 0 = disabled (fuel ratio not controlled by the autopilot) 1 = enabled
MachPoint	10	Reference Mach number for the engine control system.

i When `PA_EngineMode = 0`, the Mach-tracking loop is disabled. This does not necessarily imply that the engine is turned off.

6 TECS (TOTAL ENERGY CONTROL SYSTEM)

Simplified Total Energy Control System (TECS) configuration.

Variable	Default	Description
TECS_ON	0	Enables the TECS (set to 1).
PA_TECSEngineInput	0	Provides acceleration/deceleration reference to the TECS loop (set to 1).

i The TECS coordinates the total energy of the vehicle, coupling longitudinal motion and propulsion.

7 AUTOPILOT MODES

Select the autopilot modes and define the reference values.

Longitudinal Autopilot (PA_LongMode)	
Mode	Description
0	Disabled
1	Theta tracking
2	Altitude hold
3	Flight-path angle (gamma) tracking

Reference variables (longitudinal)

Variable	Default	Description
AltPoint	100000	Reference altitude [m]
ThetaPoint	0	Reference pitch angle [deg]
GammaRef_deg	0	Reference flight-path angle [deg]

Lateral-Directional Autopilot (PA_LatMode)	
Mode	Description
0	Disabled
1	Bank-angle tracking
2	Waypoint navigation

Reference variables (lateral)

Variable	Default	Description
PhiPoint	0	Reference bank angle [deg]
Waypoint_idx	1	Index of the active waypoint in the list
Waypoints	-	Matrix of latitude and longitude coordinates

Example waypoints (latitude,longitude):

```
Waypoints = [  
    35.6895, 139.6917;    % Tokyo  
    55.7558, 37.6173;    % Moscow  
    48.8566, 2.3522;     % Paris  
    ...  
    -8.0476, -34.8770;   % Recife  
];
```

Predefined waypoints include major world cities. Modify the matrix or select a waypoint using `Waypoint_idx`.

8 RUN A SIMULATION

Basic steps.

```
% 1. Configure options and references  
Logan_init  
  
% 2. Run the simulation  
sim('logan.slx')
```



Useful scripts (inside /Scripts)

- `Logan_init.m` - Configure model options
- `Logan_trim.m` - Trimming procedure
- `Logan_sim.m` - Example simulation script

9 FLIGHTGEAR INTEGRATION

Real-time visualization using FlightGear.

- 1 Install FlightGear (tested with v2020.3.17)
- 2 Configure the interface parameters in `Logan_init.m`
- 3 Run the simulation with the FlightGear interface enabled

10 OUTPUTS

Main simulation outputs.

- Time histories (states, control inputs, engine parameters)
- Vehicle trajectory (position, altitude, Mach number)
- Aerodynamic and propulsion performance
- Visualization in FlightGear (optional)

"Exploring the hypersonic frontier through simulation."

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